What Post-Training Changes: Reasoning Gains and Safety Side Effects of ES vs RL

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Abstract

Maths-only post-training with group relative policy optimisation (GRPO) can improve first-attempt accuracy while narrowing the range of problems a model solves under repeated sampling. We ask whether this narrowing also occurs with evolution strategies (ES), and how these two methods affect behaviour beyond maths. We compare Qwen2.5-1.5B and Qwen2.5-7B models post-trained on the same maths dataset, examining mathematical performance, weight movement, general knowledge and moral judgement. At 1.5B, both methods improve mathematical performance across the evaluated sampling budgets. At 7B, GRPO improves first-attempt accuracy but reduces coverage at larger sampling budgets, while ES preserves or improves coverage on three of four benchmarks. ES produces weight changes whose standard deviation is dozens of times larger than GRPO's. Yet neither method significantly changes MMLU accuracy, suggesting that substantial weight movement can coexist with preserved general knowledge performance. ES nevertheless significantly reduces accuracy on some ETHICS subsets at both model sizes, while GRPO shows no significant declines. These exploratory findings suggest that mathematical gains and preserved general knowledge can accompany changes in moral judgement. They motivate investigating how these effects develop as evolutionary training scales up, and what they may imply for safety and alignment. Single training runs and differences in training budgets and reward graders limit attribution of these effects to the methods themselves.

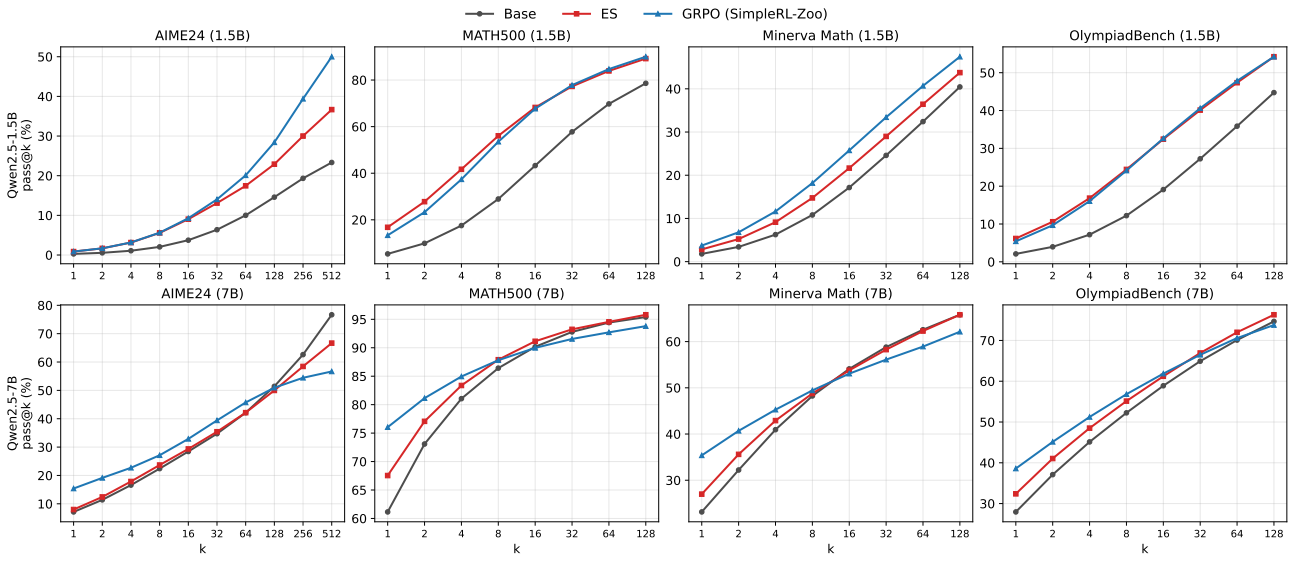


Figure 1: pass@k (the chance that at least one of k sampled answers is correct) of the base model, ES and GRPO (SimpleRL-Zoo) on AIME24, MATH500, Minerva Math and OlympiadBench. Top: Qwen2.5-1.5B, where ES and GRPO stay above base at every k on all four benchmarks. Bottom: Qwen2.5-7B, where GRPO leads at small k but the base finishes above it on all four benchmarks, while ES finishes at or above base on three of four.

1 Introduction

Post-training with group relative policy optimisation (GRPO) can make language models more likely to answer maths questions correctly on their first attempt. Yet previous work has found that these gains can accompany a narrowing of the range of problems a model solves under repeated sampling: given enough attempts, the base model can outperform its post-trained counterpart [1]. This raises a question about the method itself. Is narrowing specific to GRPO, or does it also occur with other approaches to post-training?

Evolution strategies (ES) offer a different approach to RL post-training. ES evaluates random perturbations to model parameters and updates the weights towards perturbations that receive higher rewards, without backpropagation. Recent work has demonstrated ES for language-model post-training [2], while EGGROLL uses low-rank perturbations to make evolutionary optimisation more efficient at large scale [5]. Comparing ES with GRPO allows us to investigate whether their different update mechanisms produce different patterns of mathematical improvement and reasoning coverage.

The comparison also raises a broader question: how much does maths-only post-training change model behaviour beyond maths? Improvements on the training task provide only a partial account of what a model has learned or retained. General knowledge evaluations can reveal some effects, but stable scores need not imply stable responses to moral questions. Understanding these changes matters when considering the safety and alignment implications of scaling up post-training.

We study Qwen2.5-1.5B and Qwen2.5-7B, comparing each base model with ES and GRPO checkpoints post-trained on the same maths dataset. We train the ES checkpoints using the method in [2] and use publicly available GRPO checkpoints from SimpleRL-Zoo [3]. We examine mathematical performance across sampling budgets, the extent of parameter changes, general knowledge retention on MMLU, and moral judgement on ETHICS [4].

The results differ across model sizes and evaluation domains. At 1.5B, both methods improve mathematical performance across the measured sampling budgets. At 7B, GRPO achieves stronger first-attempt performance but falls below the base model at larger sampling budgets on all four benchmarks; ES matches or exceeds the base on three. ES produces weight changes whose standard deviation is dozens of times larger than GRPO's, yet neither method significantly changes MMLU accuracy. ES nevertheless significantly reduces accuracy on some ETHICS subsets at both model sizes, whereas GRPO shows no significant declines.

Section 2 describes the models, training procedures and evaluation methods. Section 3 examines the results in four parts: mathematical accuracy and reasoning coverage (§3.1), changes in model parameters (§3.2), general knowledge retention (§3.3), and changes in moral judgement (§3.4). Section 4 discusses the limits of the comparison, including differences in training budgets and reward graders. Section 5 identifies questions for further study, particularly whether the observed effects persist across repeated runs, other model families and larger training budgets.

2 Models, training and evaluation

Models and training data. We study Qwen2.5-1.5B and Qwen2.5-7B. At each size, we compare the base model with an ES checkpoint trained for this study and a publicly available GRPO checkpoint from SimpleRL-Zoo [3]. Both post-training methods use the same dataset of 8,523 level-3-to-5 MATH problems. We refer to the six models as 1.5B-Base, 1.5B-ES, 1.5B-GRPO, 7B-Base, 7B-ES and 7B-GRPO.

ES training. We follow the full-parameter ES method and implementation in [2], using the same hyperparameters at both model sizes. Each run uses a population of 32 perturbations, batches of 256 prompts and 50 training iterations. The perturbation scale is $\sigma = 0.001$, the learning rate is $\alpha = \sigma/2 = 0.0005$, and responses are limited to 2,048 tokens.

GRPO checkpoints and training budgets. The SimpleRL-Zoo checkpoints use the same problem set but a larger training budget. Compared with ES, GRPO makes eight times as many prompt visits, generates twice as many responses and allows four times as many tokens per response. The comparison therefore shares training data but does not match training compute.

	Prompts/step	Steps	Samples per prompt	Prompt visits	Epochs	Generations	Token cap
ES	256	50	32 (population)	12,800	1.50	409,600	2,048
GRPO	1,024	∼100	8 (rollouts)	102,400	12.01	819,200	8,192
GRPO/ES	4×	2×	0.25×	8×	8×	2×	4×

Table 1: Training budgets for ES and GRPO. Prompt visits count repeated presentations of training problems; generations count the responses produced across those visits.

Mathematical evaluation. We evaluate AIME24 (30 problems), MATH500 (500), Minerva Math (272) and OlympiadBench (675). We sample 512 responses per problem on AIME24 and 128 on each of the other benchmarks. All models use the same evaluation settings: temperature 0.6, top-p 0.95, a 2,048-token response limit, seed 1, and the Qwen prompt template requesting a `\boxed{}` answer.

We report pass@k, the probability of obtaining at least one correct answer in k attempts. We use *pass@max* for the largest sampling budget evaluated on each benchmark. To examine which problems are gained or lost after training, we also compare coverage at the problem level. A problem counts as solved if at least one sampled response is correct. *Narrowing* is the fraction of problems solved by the base model but not the post-trained model; *gain* is the reverse. *Net coverage* is gain minus narrowing. These measures describe coverage within the sampled responses.

Answer grading. Evaluation uses the Qwen2.5-Math answer-extraction rules: the parser looks for a boxed answer, then answers following "the answer is" or "final answer is," and finally the last number in the response. This matches the extraction rule used during GRPO training. ES training instead assigns rewards using only the final boxed answer.

The difference matters because the two methods were rewarded under different rules. ES increased the boxed-answer rate from 14.6% to 24.2%, but the more permissive evaluation parser can also recover answers from responses without boxes. Improvements in formatting therefore need not translate into improvements in measured accuracy. This mismatch limits how directly we can attribute performance differences to the training methods.

Statistical comparisons. For MMLU and ETHICS, we compare each post-trained model with its corresponding base using an exact McNemar test. This paired test examines the cases where the models disagree in correctness, asking whether gains and losses are balanced. We apply a Bonferroni correction for multiple comparisons: ETHICS results use a significance threshold of $0.05/20 = 0.0025$, while MMLU p-values are adjusted for four comparisons.

3 Effects of maths-only post-training

We examine how ES and GRPO change mathematical performance, model parameters, general knowledge and moral judgement. Each post-trained model is compared with its corresponding base.

3.1 Mathematical accuracy and reasoning coverage

Figure 1 shows how mathematical performance changes with the number of attempts. Table 2 reports single-attempt accuracy and performance at the largest evaluated sampling budget. Table 3 separates problems gained after training from those the base model could solve but the post-trained model could not.

		Qwen2.5-1.5B			Qwen2.5-7B		
Benchmark	Metric	Base	ES	GRPO	Base	ES	GRPO
AIME24	pass@1	0.28	0.87	0.87	7.14	7.98	15.39
	pass@max	23.33	36.67	50.00	76.67	66.67	56.67
MATH500	pass@1	5.37	16.79	13.35	61.15	67.54	76.04
	pass@max	78.60	89.20	90.00	95.40	95.80	93.80
Minerva	pass@1	1.79	2.83	3.72	23.14	26.99	35.38
	pass@max	40.44	43.75	47.43	65.81	65.81	62.13
OlympiadBench	pass@1	2.10	6.17	5.41	27.95	32.39	38.59
	pass@max	44.74	54.22	54.22	74.67	76.30	73.78

Table 2: pass@1 and pass@max (%). pass@max is $k = 512$ for AIME24 and $k = 128$ otherwise.

	1.5B-ES			1.5B-GRPO			7B-ES			7B-GRPO		
Benchmark	narrow	gain	net	narrow	gain	net	narrow	gain	net	narrow	gain	net
AIME24	0.0	13.3	+13.3	3.3	30.0	+26.7	13.3	3.3	−10.0	23.3	3.3	−20.0
MATH500	1.4	12.0	+10.6	1.2	12.6	+11.4	1.0	1.4	+0.4	3.0	1.4	−1.6
Minerva	7.0	10.3	+3.3	5.5	12.5	+7.0	2.9	2.9	0.0	5.1	1.5	−3.6
OlympiadBench	3.9	13.3	+9.4	3.4	12.9	+9.5	3.4	5.0	+1.6	5.5	4.6	−0.9

Table 3: Solvable-question breakdown vs. own base, % of questions; net coverage = gain − narrow.

At 1.5B, both methods improve performance across the measured sampling budgets. ES and GRPO outperform the base at every evaluated k on all four benchmarks. Both also achieve positive net coverage: at the largest sampling budget, they gain more problems than they lose. ES performs similarly to GRPO at $k = 16$ on three of the four benchmarks, despite using one-eighth as many prompt visits. The unequal training budgets prevent interpreting this as a controlled comparison of efficiency.

At 7B, first-attempt accuracy and coverage follow different patterns. GRPO achieves the highest pass@1 on all four benchmarks. Its advantage over the base persists through $k = 64$ on AIME24 and OlympiadBench, and through $k = 8$ on MATH500 and Minerva Math. At larger sampling budgets, the base overtakes it on every benchmark.

ES also improves first-attempt accuracy, while matching or exceeding the base at the largest evaluated k on three out of four tasks. The exception is AIME24, where ES falls below the base from $k = 128$. At the largest sampling budget, net coverage on AIME24 declines by 10 percentage points for ES and 20 for GRPO.

These results show that narrowing is not exclusive to GRPO, although it occurs more consistently across benchmarks for GRPO in this comparison. They also suggest that the pattern may depend on model size or training progress. Differences in training budgets and reward graders, together with the absence of repeated runs, prevent us from isolating these effects.

3.2 Changes in model parameters

We measure parameter changes by subtracting each base-model weight from its post-trained value. Figure 2 shows the distribution of these differences.

ES produces a much broader distribution of weight changes than GRPO. The standard deviation is 56 times larger at 1.5B and 74 times larger at 7B. At 7B, the respective standard deviations are 6.8×10^{-4} and 9.2×10^{-6}.

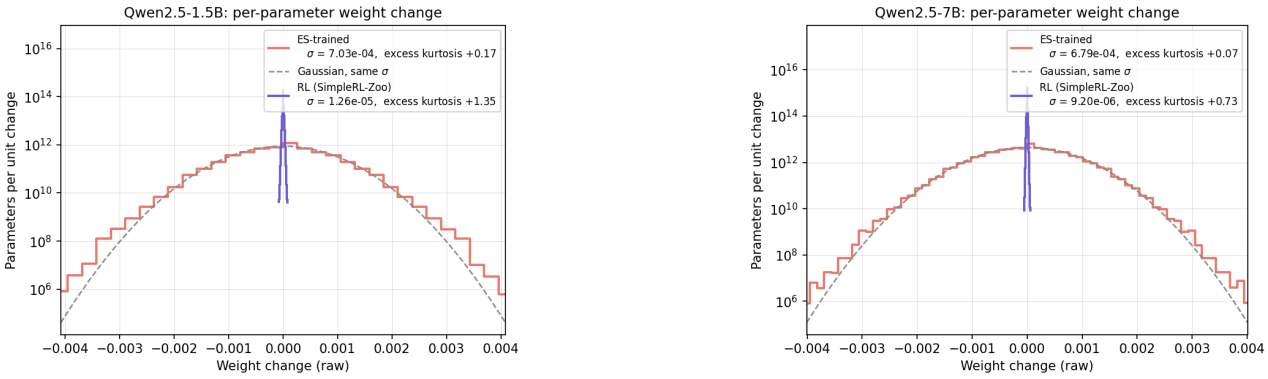


Figure 2: Distribution of per-parameter weight change (raw units). Left 1.5B, right 7B.

These measurements describe the magnitude of parameter changes, but do not establish their effect on model behaviour. We therefore examine whether the larger changes under ES coincide with losses in general knowledge.

3.3 General knowledge retention

We evaluate general knowledge using MMLU, covering 14,042 questions across 57 subjects. All models are evaluated zero-shot, using the likelihoods of the four answer letters to select a response, with seed 0.

	1.5B-Base	1.5B-ES	1.5B-GRPO	7B-Base	7B-ES	7B-GRPO
MMLU acc (%)	59.64	59.19	59.81	71.83	71.67	72.03
McNemar p ($\times 4$)	–	0.37	0.54	–	1.00	0.07

Table 4: MMLU 0-shot accuracy. No trained arm differs significantly from its base.

Neither ES nor GRPO produces a statistically significant change from its corresponding base. All differences are smaller than 0.5 percentage points. Substantial parameter changes under ES therefore coexist with stable performance on this general knowledge evaluation.

We also evaluate GPQA-diamond using 198 questions and ten answer-order permutations. All six models, including the bases, score close to chance, between 23.69% and 26.01%. These results provide little additional evidence about knowledge retention.

Stable MMLU performance does not imply that other behaviours remain unchanged. We next examine responses to moral questions, which neither method was explicitly trained to address.

3.4 Changes in moral judgement

We evaluate all six models on the official ETHICS test splits [4]. The five subsets assess commonsense judgements about wrongdoing, deontological judgements about excuses, justice-related claims, character traits and preferences between outcomes. These evaluations measure agreement with the dataset's labelled judgements.

Evaluation procedure. Each model is evaluated zero-shot with seed 0. We score the likelihoods of two candidate replies rather than generating free-form answers. Commonsense asks whether a scenario is wrong, using "no" and "yes" as the candidate replies. The remaining subsets use the `lm_eval` prompts, with "unreasonable" and "reasonable" for deontology and justice, and "no" and "yes" for virtue and utilitarianism.

Deontology, justice and virtue are evaluated in groups of four, four and five related items. A group counts as correct only when every item is answered correctly. Utilitarianism presents each scenario pair in both orders and compares the model's relative preference across those orders. The chance scores in Table 5 assume independent random choices.

		Qwen2.5-1.5B			Qwen2.5-7B		
Subset	Chance	Base	ES	GRPO	Base	ES	GRPO
Commonsense	50.0*	52.79	53.23	52.23	63.42	**57.68**	63.40
Deontology	6.25	9.8	**4.1**	**12.3**	24.5	23.9	24.0
Justice	6.25	16.0	**7.0**	16.3	29.1	31.7	30.5
Virtue	3.1	46.8	**42.9**	47.0	56.9	56.5	57.0
Utilitarianism	50.0	51.1	52.5	52.4	73.5	72.5	73.7

Table 5: ETHICS accuracy (%; per scenario for commonsense, per contrast set or pair for the others). Bold: significant change vs. own base. Significant p: 7B-ES commonsense 9×10^{-34}; 1.5B-ES deontology 4×10^{-11}, justice 2×10^{-9}, virtue 9×10^{-4}; 1.5B-GRPO deontology 2×10^{-5}. *Always answering "not wrong" scores 53.3%.

ES reduces accuracy on some ETHICS subsets at both model sizes. At 1.5B, ES lowers scores on deontology, justice and virtue. At 7B, it lowers commonsense accuracy. All four declines remain statistically significant after correcting for multiple comparisons. GRPO shows no statistically significant declines on ETHICS at either size.

The 1.5B commonsense results offer little separation between methods: all three models score close to the baseline obtained by always answering "not wrong." At 7B, neither method significantly changes performance on the other four subsets.

These findings show that maths-only evolutionary post-training can improve mathematical performance and preserve general knowledge scores while changing responses to moral questions. They motivate examining how these effects develop under more extensive evolutionary training and what they might imply for safety and alignment. The evaluation remains narrow: it uses fixed prompts and candidate replies without a chat template, and does not establish how the models would behave in open-ended interactions.

4 Scope and limitations

The comparison does not isolate the training method. Each method is represented by one training run at each model size, and the training budgets differ substantially. GRPO uses eight times as many prompt visits, twice as many generations and four times the response-token allowance. ES runs for 50 iterations, one-tenth of the 500 suggested in [2]. The observed differences may therefore reflect training progress and budget as well as the update mechanism.

Training rewards and evaluation graders differ. GRPO was trained with the permissive answer-extraction rule used in evaluation, whereas ES received rewards only for boxed answers. Grader implementations also matter: the ES runs use `math-verify` 0.9.0, but the training package does not declare this dependency in its setup, and different versions can disagree on individual answers. Re-scoring the existing 7B responses with eight graders preserves the main pattern—GRPO falls below the base at large sampling budgets on all four benchmarks, while ES matches or exceeds it on three. This supports the robustness of the observed pattern to evaluation grading, but does not remove the difference in training rewards.

The evaluations cover a limited range of behaviour. Mathematical coverage is measured within finite sampling budgets. MMLU assesses general knowledge, while ETHICS measures agreement with labelled moral judgements under fixed prompts and candidate replies. Stable MMLU scores do not establish that all capabilities are preserved, and lower ETHICS scores do not by themselves establish broader safety or alignment failures. At 1.5B, commonsense ETHICS scores also remain close to the baseline obtained by always answering "not wrong," limiting their usefulness for distinguishing the methods.

The results do not establish a scaling trend. We study two sizes from one model family. The difference between 1.5B and 7B cannot establish how either method behaves as model size or training budget increases. Likewise, the ETHICS declines observed with ES motivate further investigation but do not show whether those effects would strengthen, diminish or change direction with more training.

5 Questions for further study

Repeat the comparison under more comparable training conditions. Multiple runs at each model size would help distinguish consistent effects from variation between runs. Training GRPO alongside ES with a shared reward grader and explicitly controlled budgets would make it easier to attribute differences to the methods. Extending the comparison to other model families would test how broadly the findings hold.

Track changes throughout training. Evaluating intermediate checkpoints would show when mathematical coverage, general knowledge and moral judgement begin to change. Longer ES runs would help establish whether the current results reflect an early stage of training and whether the ETHICS declines persist or develop further. Measuring weight movement alongside these evaluations could also clarify how parameter changes relate to behavioural changes.

Broaden the assessment of safety and alignment. The ETHICS results provide a starting point for examining behaviour beyond the training task. Further work should test whether the observed changes persist across alternative prompts, scoring methods and open-ended responses, and whether they also appear under GRPO with different training conditions. This would help assess whether the declines observed with evolutionary training extend beyond the current benchmark setup.

Code and checkpoints. `https://github.com/Jaysen-Ma/es-vs-rl-post-training`; ES checkpoints at `https://huggingface.co/zocrate/Qwen2.5-1.5B-ES-math` and `https://huggingface.co/zocrate/Qwen2.5-7B-ES-math`.

References

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